Multistage Enhancement Network for Tiny Object Detection in Remote Sensing Images

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***Abstract*— With the rapid advances in deep learning techniques, remote sensing object detection (RSOD) has achieved remarkable achievements in recent years. However, tiny object detection remains unsatisfactory and suffers from two main drawbacks: 1) the high sensitivity of IoU for location deviation in tiny objects and 2) the poor-quality feature representations of tiny objects. To address the aforementioned problems, we propose a multistage enhancement network (MENet) that achieves the instance-level and feature-level enhancement of tiny objects from different stages of the detector. Since the IoU-based label assignment drastically deteriorates the positive samples for tiny objects, we first propose a central region (CR)-based label assignment to substitute it in the region proposal network (RPN). The CR label assignment regards the anchors that fall into the CR of ground-truth boxes as positive samples, which provides more positive samples for tiny objects. Then, we design a gated context aggregation (GCA) module that selectively aggregates valuable context information to enhance the feature representation of tiny objects. Additionally, we devise a positive RoI (pRoI) feature generator in the region convolutional neural network (R-CNN) to generate a rich diversity of high-quality pRoI features for tiny objects. We conduct extensive experiments on AI-TOD and SODA-A datasets, and the results demonstrate the effectiveness of our proposed method.**

***Index Terms*— Context aggregation, label assignment, remote sensing images (RSIs), tiny object detection.**

I. Introduction

REMOTE sensing object detection (RSOD) aims to accurately localize and identify objects of interest in the given remote sensing images (RSIs). As a fundamental yet important task in RSI interpretation, RSOD has wide applications in precision agriculture, urban planning, and wildlife conservation. In recent years, the swift advancement of Earth observation technologies has propelled human beings into an era of remote sensing big data and pushed RSOD to hot-spot research with unprecedented attention [1], [2], [3]. Thanks to the powerful feature representation capabilities of deep learning techniques, the deep learning-based detectors have demonstrated superior detection performance in the natural scene. Thus, many researchers have extended the deep learning-based detectors into the remote sensing community and presented various methods to overcome challenges such as scale variations [4], [5], arbitrary orientations [6], [7], background interference [8], [9], etc., leading to remarkable breakthroughs in RSOD. However, this prosperity could not camouflage the deficiencies of tiny object detection.



Fig. 1. (Left) Tiny objects typically exhibit limited scales and poor-quality feature representations. The IoU-based label assignment restricts the number of positive samples for tiny objects in both (top-right) RPN and (bottom-right) R-CNN.

Actually, tiny objects are prevalent in RSIs. On the one hand, the typical resolution for RSIs is 1–3 m, which means that objects in most categories (e.g., airplanes, ships, and storage tanks) may appear less than 16 × 16 pixels. On the other hand, even in a high-resolution RSI with a ground sampling distance of 0.3 m/pixel, vehicles can only cover 10 × 10 pixels, as illustrated in Fig. 1. Such limited scales of tiny objects result in poor-quality feature representations, further increasing the challenges of tiny object detection.

Recent studies [10], [11], [12] pointed out that tiny objects are sensitive to the IoU metric. Even a slight location deviation can result in severe IoU variations. This unstable change makes the region proposal network (RPN) fail to assign sufficient positive samples for tiny objects under the IoU-based label assignment (as shown in the top-right of Fig. 1) and severely deteriorates the tiny object detection performance. Therefore, it is crucial to increase the number of positive samples for tiny objects in RPN. Moreover, the region convolutional neural network (R-CNN) in the two-stage detector also employs the IoU-based label assignment to select the positive and negative RoI features, limiting the number of positive RoI (pRoI) features for tiny objects, as depicted in the bottom-right of Fig. 1. Apart from the IoU-sensitive problem, the tiny object also suffers from the challenge of poor-quality feature representations. The well-known feature pyramid network (FPN) and its extensions [13], [14], [15] have shown effectiveness

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in improving the feature representations for tiny objects. However, they mainly contain local information and lack the learning of context information that can provide auxiliary information to tiny objects.

Drawing on the above analysis, we propose a multistage enhancement network (MENet) for tiny object detection in RSIs. The proposed MENet achieves instance-level and feature-level enhancement of tiny objects from the different stages (i.e., RPN, FPN, and R-CNN) of the detector and significantly improves the tiny object detection performance. First, we propose a central region (CR)-based label assignment to improve the number of positive samples for tiny objects in RPN, where the anchors are assigned as the positive samples if they are inside the CR of the ground-truth box. The proposed CR label assignment addresses the limitation of the IoU-based label assignment on tiny objects and provides more positive samples for tiny objects in RPN. Second, we propose a gated context aggregation (GCA) module based on FPN to enhance the feature representation of tiny objects. On the one hand, we employ an efficient attention mechanism (EAM) to capture the context information for tiny objects. On the other hand, we design a gated fusion mechanism to selectively aggregate valuable context information and ignore the interference of massive background information. Third, we note that previous works only consider the negative impact of IoU-based label assignment for tiny objects in the RPN while ignoring its hindrance to the R-CNN. Thus, we propose a pRoI feature generator to generate more high-quality RoI features for tiny objects in the R-CNN.

Our main contributions can be summarized as follows.

1) We propose a MENet that adopts a multistage enhancement approach to improve tiny object detection from the instance level (positive samples of tiny objects) and the feature level (feature representations of tiny objects).
2) We design a CR label assignment to replace the IoU-based label assignment in RPN, facilitating the positive samples assignment for tiny objects.
3) We present a GCA module to selectively aggregate valuable context information to improve the feature representation of tiny objects.
4) We devise a pRoI generator to generate a rich diversity of pRoI features for tiny objects in the R-CNN.

The remainder of this article is organized as follows. In Section II, we provide a brief review of the related work. Then, we elaborate on our proposed modules in Section III. Section IV reports the extensive experiments in two datasets. Finally, Section V concludes this article.

II. Related Works

A. Object Detection in RSIs

The past two decades have witnessed the flourishing of object detection in RSIs. The early stage methods mainly adopted hand-crafted features or prior knowledge, yielding limited performance [16], [17], [18]. Later, machine learning approaches became mainstream in RSOD [19], [20], [21]. However, these shallow learning-based features from machine learning approaches still lack deep semantic information, restricting the discriminative representations of objects.

Recently, deep learning-based detectors have made remarkable achievements in computer vision [22], [23], [24]. Along with the advanced development, deep learning-based detectors have dominated the RSOD field and achieved tremendous breakthroughs [1].

Due to the diverse spatial resolutions of RSIs, scale variation is one of the notorious challenges in RSOD and has received widespread attention [25], [26], [27]. FSOD-Net [25] introduces three scale-invariant regression layers to capture full-scale object information. Fu et al. [26] designed a feature-fusion architecture that extracts multi-scale features with detailed spatial and rich semantic information. The overhead shooting views lead to arbitrary orientations of remote sensing objects, and rotated object detection in RSIs has garnered significant interest [28], [29], [30], [31]. Han et al. [30] proposed the S^2ANet that encodes the rotation information in the network to acquire rotation equivariance and rotation invariance features. Yao et al. [31] designed a simple yet effective rotated object representation under the polar coordinate system for handling the boundary discontinuity problem. Considering the massive background noise in RSIs, some researchers highlight the foreground region by explicitly injecting the foreground information into the detector [32], [33], [34]. Moreover, recent studies have attempted to perform RSOD in limited supervision scenarios (e.g., weakly supervised learning [35], [36], [37] and few-shot learning [38], [39], [40]) to alleviate the time-consuming and labor-intensive annotation task.

Previous works put major attention to the typical RSOD challenges of scale variation, arbitrary orientations, and background interference while ignoring the tiny scales of geospatial objects. In practice, tiny objects are widely distributed in RSIs and suffer from unsatisfactory detection performance. Thus, this article concentrates on tiny object detection in RSIs and proposes a MENet to enhance the tiny object detection performance.

B. Tiny Object Detection in RSIs

Current studies on tiny object detection in RSIs can be categorized into three main groups: discriminative feature learning, super-resolution-based methods, and improved detection strategies.

1) Discriminative Feature Learning: Tiny objects typically exhibit weak feature responses, which increases the challenges of learning their features. In response, many methods adopt discriminative feature learning to improve the tiny object detection performance [41], [42], [43], [44]. Li et al. [41] presented cross-layer attention that integrates high-level semantic information into shallow features to obtain stronger features of tiny objects. Wang et al. [42] leveraged the atrous spatial feature pyramids (ASFPs) to extract surrounding context information that compensates for weak feature responses of tiny objects. Huang et al. [43] fused the multiscale context features into tiny objects to enhance the feature response of tiny objects.

2) Super-Resolution-Based Methods: The super-resolution-based methods claim that the tiny scale is the critical problem in tiny object detection, so these methods integrate super-resolution modules into the tiny object detection pipeline. Rabbi et al. [45] proposed an edge-enhanced super-resolution generative adversarial network (GAN) to enhance the visual details of tiny objects. Wu and Xu [46] designed a

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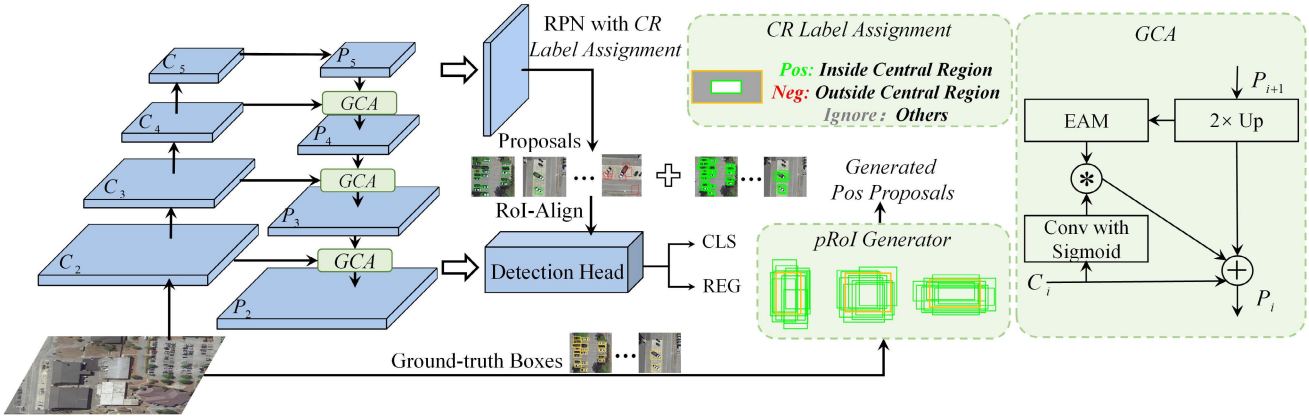


Fig. 2. Overall framework of our proposed MENet. The MENet adopts a multistage enhancement approach to improve tiny object detection. In RPN, the proposed CR label assignment replaces the IoU-based label assignment, providing more positive samples for tiny objects. The proposed GCA module is integrated into FPN to improve the feature representation of tiny objects. In the R-CNN, a pRoI generator is designed to generate a rich diversity of pRoI features for tiny objects.

two-step tiny object detection framework, where the method first obtains proposal regions based on the key-point detection and then employs a multi-task GAN to perform the super-resolution and detection tasks on these proposal regions. However, the direct image-level super-resolution imposes extra computational complexity. Thus, some studies [47], [48] adapted the super-resolution strategy at the feature level to effectively save computational resources.

3) Improved Detection Strategies: Different from the above two streams of methods, the recent work points out that the prevailing detection paradigms are improper for tiny object detection and inevitably degrade their performance. Pang et al. [49] highlighted that the extensive pooling operations cause tiny objects to disappear from feature maps and proposed a zoom-out and zoom-in architecture to improve the recall of tiny objects. Wu et al. [11] found that the misalignment of multiscale feature fusion restricted tiny object detection performance and proposed a feature-aware alignment module to alternative this problem. Meanwhile, the IoU-based label assignment is sensitive to the location deviation in tiny objects, leading to insufficient positive samples for tiny objects. Xu et al. [10] adopted the 2-D Gaussian distributions to represent the objects and designed the normalized Wasserstein distance (NWD) to measure the similarity of Gaussian distributions. Fu et al. [12] proposed a Gaussian similarity-based adaptive dynamic label assignment for improving tiny object detection.

Inspired by previous works, our proposed MENet enhances the tiny objects at both instance and feature levels from three complementary viewpoints.

III. Proposed Method

Fig. 2 shows the overall structure of our proposed MENet. We adopt the anchor-based faster R-CNN as the baseline and introduce CR label assignment, GCA module, and pRoI generator to different stages of the baseline to improve the tiny object detection performance.

A. CR-Based Label Assignment

Since the tiny objects exhibit severe IoU variations for slight location deviation, the RPN struggles to assign sufficient positive samples for tiny objects under the IoU-based label assignment, seriously limiting the learning of tiny objects.

To this end, we propose the CR label assignment that assigns positive and negative samples according to different regions of the ground-truth box. Algorithm 1 describes the details of our proposed method. For each ground-truth box $g \in G$, we first compute its CR. Then, we assign the anchors whose center point falls into the CR as candidate positive samples. We consider that the anchors falling into the CR of the ground-truth box are more likely to be positive samples and contain more relevant information about the ground-truth box. However, the above strategy only considers the location

Algorithm 1 CR-Based Label Assignment

Input:
G is a set of ground-truth boxes on the image
$\mathcal{A}$ is a set of anchors
L is the long edge of the central region
Output:
$\mathcal{P}$ is a set of positive samples
$\mathcal{N}$ is a set of negative samples
$\mathcal{I}$ is a set of ignored samples

for each ground-truth box $g \in G$ **do**
  compute the central region of the ground-truth box g, according to the L;
  select anchors whose center point falls into the central region as the candidate positive samples $\mathcal{C}_g$;
  compute IoU between $\mathcal{C}_g$ and g: $D_g = IoU(\mathcal{C}_g, g)$;
  compute mean of D_g as threshold: $\theta_g = Mean(D_g)$;
  for each candidate positive samples $c \in \mathcal{C}_g$ **do**
    if $IoU(c, g) \geq \theta_g$ **then**
      $\mathcal{P} = \mathcal{P} \cup c$
    end if
  end for
end for
select anchors that do not fall inside any ground-truth box as the negative samples $\mathcal{N}$;
$\mathcal{I} = \mathcal{A} - \mathcal{P} - \mathcal{N}$
return $\mathcal{P}, \mathcal{N}, \mathcal{I}$

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relationship while ignoring the scale consistency between the anchors and the ground-truth box, which may introduce noisy positive samples. Inspired by ATSS [50], we calculate the IoU between all candidate positive samples and ground-truth box g and use the mean value as a demarcation threshold θ_g. The candidate positive samples with an IoU larger than θ_g are assigned as the final positive samples. Based on this statistical characteristic of each ground-truth box, we can select positive samples from more suitable feature pyramid levels. Besides, if an anchor is selected as the positive sample for multiple ground-truth boxes, we assign it to the ground-truth box with a larger IoU. For anchors that do not fall inside any ground-truth box, we assign them as negative samples since they contain more information outside the ground-truth box. The rest anchors are assigned as the ignored samples, which are not involved in the network training phase.

The above program shows that the CR is vital to the CR label assignment and directly affects the number of positive samples assigned to each ground-truth box. Thus, we devise a fixed area CR selection method. On the one hand, the fixed area CR can provide more positive samples for tiny objects than the adaptive area. On the other hand, it alleviates the imbalance of positive samples across objects of different scales. Concretely, if the long edge of a ground-truth box $g = (x, y, w, h)$ is greater than the predefined threshold L, we define the region with an area of $L \times (L/\mathrm{AR}_g)$ and centered on (x, y) as the CR of ground-truth box, where AR_g stands for the aspect ratio of the ground-truth box g and the long edge of the CR is parallel to the long edge of the ground-truth box. The CR keeps the same aspect ratio as the ground-truth box, facilitating the positive sample assignment, especially for objects with large aspect ratios in the RSI (e.g., bridges, harbors, etc.). Besides, if the long edge of a ground-truth box is smaller than the predefined threshold L, we assign all anchors that fall into the ground-truth box as the positive samples, which can provide more positive samples for tiny objects.

B. Gated Context Aggregation

The limited scale of tiny objects leads to poor-quality feature representation and seriously affects the detection performance. FPN iteratively propagates rich semantic information to the lower level features and improves the feature representation of tiny objects [13]. Given the multilevel features from the backbone, the FPN produces the semantic-enhanced feature pyramid by aggregating the same level backbone feature $C_i \in R^{256\times H_i \times W_i}$ with the higher level FPN feature $P_{i+1} \in R^{256\times H_{i+1}\times W_{i+1}}$, which can be expressed as follows[1]:

$$P_i = \mathrm{Conv}_{f_i}(C_i + U(P_{i+1})) \tag{1}$$

where $U(\cdot)$ denotes the up-sampling operation with a factor of 2, and Conv_{f_i} defines the 3×3 convolution layer.

However, the FPN remains to learn local information and lacks capturing context information. Thus, we propose a GCA module to selectively aggregate valuable context information for tiny objects while ignoring irrelevant background noise.

1) Efficient Attention Mechanism: Considering the poor-quality feature representation of tiny objects, we aim to introduce the context information to assist the feature representation of tiny objects. The self-attention mechanism (SAM) has demonstrated its effectiveness in modeling long-range interaction [51]. As shown in (2), the SAM establishes long-range dependency by measuring the similarity between all pairs of positions on the feature $x \in R^{N\times d}$

$$S(Q, K, V) = \mathrm{Softmax}_{\mathrm{row}}\left(QK^T\right)V \tag{2}$$

where $Q = xW_q, K = xW_k, V = xW_v$ represents embedding features and $W_{q,k,v} \in R^{d\times d}$ is the learnable matrix. $\mathrm{Softmax}_{\mathrm{row}}(\cdot)$ denotes applying the softmax function along each row of the matrix. However, the SAM suffers from a secondary computational complexity of $O(N^2)$, which incurs high memory and computational resources to process high-resolution feature maps (e.g., P_2, P_3 level features in FPN).

To alleviate the above computational burden, we introduce an EAM with a linear complexity of $O(N)$ [52], and the structure is presented in Fig. 3. Unlike the SAM that first measures the pairwise similarities between all points, the proposed EAM regards the normalized key features K as d global semantic attention maps of the input features and aggregates the value features V through weighted summation to form global context vectors G. This global context information is then multiplied by the normalized Q to obtain the final result. Mathematically, the EAM is described as follows:

$$E(Q, K, V) = \mathrm{Softmax}_{\mathrm{row}}(Q)\left(\mathrm{Softmax}_{\mathrm{col}}(K)^T V\right). \tag{3}$$

Following the combination property of matrix multiplication, the EAM is mathematically equivalent to the SAM but effectively reduces the computational complexity. Compared to the other extensions of SAM [53], [54], the proposed EAM provides a more concise way to reduce the computational complexity of the SAM.

2) Gated Fusion Mechanism: Since the remote sensing objects are always exhibited in complex backgrounds, we design a gated fusion mechanism to selectively aggregate valuable context information into low-level features. In practice, we apply a 1×1 convolution layer followed by a sigmoid function to the low-level feature C_i to generate the spatial attention map $W_i \in R^{1\times H_i \times W_i}$, which is then used to guide the context aggregation to low-level features.

By embedding the EAM and the gated fusion mechanism into FPN, we get the proposed GCA module, which is described as follows:

$$\begin{aligned} \mathrm{Context}_{i+1} &= W_i * E(Q_{i+1}, K_{i+1}, V_{i+1}) \\ P_i &= \mathrm{Conv}_{f_i}(C_i + U(P_{i+1}) + \mathrm{Context}_{i+1}) \end{aligned} \tag{4}$$

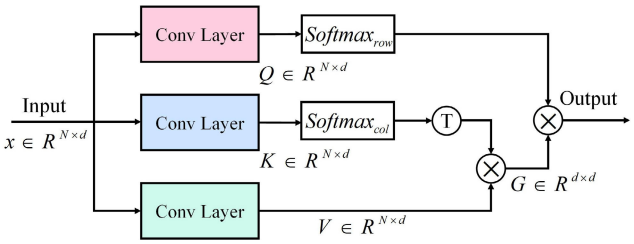


Fig. 3. Illustration of the structure of EAM, where Q, K, and V stand for the embedding features of the input feature x. The EAM first aggregates the normalized key features K with the value features V to form global context vectors G. Then, the global context vectors are multiplied by the normalized Q to obtain the final result.

[1]Here, we simplify the expression of FPN and only depict the P_2, P_3, P_4 level features while omitting the P_5, P_6 level features in FPN.

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where $Q_{i+1}, K_{i+1}, V_{i+1}$ denotes three embedding features of $U(P_{i+1})$. Compared to the FPN, the proposed GCA module selectively aggregates valuable context information to low-level features.

C. pRoI Feature Generator

Existing two-stage detectors also rely on the IoU-based label assignment to assign the positive/negative labels for RoI features in the R-CNN. Due to the high sensitivity of IoU to tiny objects' location deviations, the IoU-based label assignment degrades the diversity of pRoI features, which hinders the accuracy of classification and regression. To address the above issue, we design a pRoI generator to directly generate rich pRoI features around each ground-truth box. With such a design, we can escape the limitations of the IoU-based label assignment and improve the diversity of pRoI features, especially for tiny objects.

Concretely, given an image with K ground-truth boxes $G = \{(x_k, y_k, w_k, h_k)\}_{k=1}^{K}$, where (x_k, y_k) and (w_k, h_k) denote the center point and two edges of the kth ground-truth box, we can generate a series of proposals $\mathcal{P}_G$ by perturbing the location of ground-truth boxes. The generated proposals can be formulated as follows:

$$\begin{aligned} \mathcal{P}_G &= \cup_{k=1}^{K} \mathcal{P}_k \\ \mathcal{P}_k &= \{p_m \mid p_m = g_k \times (1 + \delta s_k), m = 1, \cdots, M\} \end{aligned} \tag{5}$$

where $g_k = (x_k, y_k, w_k, h_k)$ and $s_k = (w_k, h_k, w_k, h_k)$ denote the kth ground-truth box and its scale coefficient. $\delta \sim \mathbb{N}(\mu, \sigma^2)$ represents the offset coefficient. We set $\mu = 0$ and $\sigma = 1/6$ to maintain the generated proposals are positive (IoU > 0.5). The Gaussian distribution has the advantage of providing more high-quality proposals, which can remedy the problem that the proposals predicted by RPN are mainly from low IoU levels [55].

Based on the generated proposals, we can extract the corresponding pRoI features by employing the RoI-Align operation [56]. Besides, we modulate the loss weight of generated RoI features to prevent the original RoI features from being overwhelmed by the generated ones.

D. Loss Function

The loss function of the proposed MENet consists of the RPN loss and R-CNN loss. The RPN loss uses the form of multitask loss following faster R-CNN [22], and it is defined as:

$$\begin{aligned} \mathcal{L}_{\text{rpn}} = \frac{1}{N_{\text{anchor}}} \Bigg(& \sum_{i=1}^{N_{\text{anchor}}} \mathcal{L}_{\text{cls}}\left(p_i, \left[l_i^* \geq 1\right]\right) \\ & + \sum_{i=1}^{N_{\text{anchor}}} \left[l_i^* \geq 1\right] \mathcal{L}_{\text{reg}}\left(t_i, t_i^*\right) \Bigg) \end{aligned} \tag{6}$$

where N_{anchor} represents the number of anchors. p_i is the predicted probability over foreground and background classes, and l_i^* is the ground-truth label. $[l_i^* \geq 1]$ is the Iverson bracket indicator function. t_i and t_i^* stand for the predicted regression offset and the ground-truth regression offset in the RPN. Following the faster R-CNN [22], $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{reg}}$ represent the cross-entropy loss and smooth-$L1$ loss.

As for the R-CNN loss, it also uses the form of multitask loss, which is expressed as follows:

$$\mathcal{L}_{\text{rcnn}} = \frac{1}{N_{\text{RoI}}} \left(\sum_{j=1}^{N_{\text{RoI}}} \lambda_j \mathcal{L}_{\text{cls}}\left(c_j, l_j^*\right) + \sum_{j=1}^{N_{\text{RoI}}} \left[l_j^* \geq 1\right] \lambda_j \mathcal{L}_{\text{reg}}\left(t_j, t_j^*\right) \right) \tag{7}$$

where N_{RoI} represents the total number of original RoI features and generated RoI features. c_j is the predicted probability over all classes, and l_j^* is the ground-truth label. t_j and t_j^* stand for the predicted regression offset and the ground-truth regression offset in the R-CNN. $\lambda_j = 1.0$ for the original RoI features and $\lambda_j = 0.1$ for the generated RoI features.

Hence, the loss function of the proposed MENet can be written as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{rcnn}}. \tag{8}$$

IV. Experiments

A. Datasets

1) AI-TOD [57]: The AI-TOD dataset is specifically designed for tiny object detection in RSIs. The dataset contains 28 036 images with a size of 800 × 800 and 700 621 annotated objects. The objects in this dataset have an average scale of 12.8 pixels, and 85.6% of them are smaller than 16 pixels, increasing the difficulties of this dataset. There are eight categories, including airplane (AI), bridge (BR), storage-tank (ST), ship (SH), swimming-pool (SP), vehicle (VE), person (PE), and windmill (WM). The whole dataset is randomly divided into three subsets, where 2/5 for training, 1/10 for validation, and 1/2 for testing. We adopt the training and validation sets for training and evaluate the performance on the test set.

2) SODA-A [58]: The SODA-A dataset is a challenging rotated tiny object detection dataset. The dataset consists of 2513 images and 872 069 annotated objects across nine categories, including airplane (AI), helicopter (HC), small-vehicle (SV), large-vehicle (LV), ship (SH), container (CO), storage tank (ST), swimming pool (SP), and windmill (WM). The whole dataset is split into training, validation, and test sets, and each subset occupies approximately 50%, 20%, and 30% approximately. In our experiment, we use the training set for training and the test set for inference.

B. Implementation Details

Our proposed MENet is based on faster-RCNN [22] and adopts the pretrained ResNet-50 [59] with an FPN [13] as the backbone. In the training phase, we train the detector for a total of 12 epochs, where the learning rate starts from 0.01 and decreases to 0.001 and 0.0001 at the 8th and 11th epochs. We also implement the linear warm-up strategy with a learning ratio of 0.333 for the first 500 iterations. The stochastic gradient descent (SGD) method with a weight decay of 5e−4 and a momentum of 0.9 is employed as the optimizer. The training batch size is set to 8 on four NVIDIA GeForce GTX 1080Ti GPUs. In the inference phase, we set the score and the IoU threshold to 0.05 and 0.5 in the non-maximum suppression (NMS).

Besides, there are slight differences in the implementation details for each dataset, considering the characteristics of

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TABLE I

ABLATION STUDY OF EACH PROPOSED MODULE IN MENET ON THE AI-TOD TEST SET. "CR": CR-BASED LABEL ASSIGNMENT. "GCA": GATED CONTEXT AGGREGATION "pRoI": POSITIVE RoI FEATURES GENERATOR

CR	GCA	pRoI	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
			11.6	26.9	8.3	0.1	7.8	**24.4**	**34.1**
✓			19.0	47.9	11.3	7.5	19.9	22.8	29.0
✓	✓		19.4	48.7	11.3	8.0	19.9	22.6	30.1
✓		✓	20.0	48.9	12.9	8.4	21.1	23.1	29.6
✓	✓	✓	**20.4**	**50.0**	**12.9**	**8.9**	**21.4**	23.2	31.0

each dataset. For the AI-TOD dataset, we preserve the top 3000 scored predicted bounding boxes per image following the benchmark.[2] For the SODA-A dataset, the original images are cropped into 800 × 800 patches with a gap of 150 pixels, and the image size is kept at 800 × 800 in both the training and inference phases. We provide the top 2000 scored predicted bounding boxes per image following the official code library.[3] Furthermore, since the SODA-A dataset has rotated annotations, the minimum bounding rectangle of the rotated ground-truth box is utilized when applying the proposed CR label assignment and pRoI generator to the rotated faster R-CNN [22] and RoI-transformer [28].

C. Evaluation Metrics

We employ the average precision (AP) metric [60] to evaluate the detection performance. For the AI-TOD dataset, we report the main seven detection metrics following the AI-TOD evaluation protocols [57]. Specifically, AP measures the mean value of AP under the IoU threshold from 0.5 to 0.95 with intervals of 0.05. AP_{50} and AP_{75} indicate the AP under the IoU thresholds of 0.5 and 0.75. Moreover, AP_{vt}, AP_t, AP_s, and AP_m are used to evaluate the performance of objects at different scale ranges, where *very tiny*, *tiny*, *small*, and *medium* denote the scale ranges of [2, 8], [8, 16], [16, 32], and [32, 64], respectively. Following the SODA-A benchmark [58], we evaluate the detection performance with AP metric in four object scale subsets: *Small* [0, 32], *extremely Small* [0, 12], *relatively Small* [12, 20], and *generally Small* [20, 32].

D. Ablation Study

In this section, we perform an extensive ablation analysis to better understand the effectiveness of our proposed modules. All experiments are evaluated on the AI-TOD test set.

1) Effectiveness of CR-Based Label Assignment: The IoU-sensitivity problem of tiny objects makes the RPN fail to assign sufficient positive samples for tiny objects under the IoU-based label assignment. To tackle this problem, we design the CR label assignment to replace the IoU-based label assignment in RPN. As shown in Table I, our proposed CR label assignment improves the AP, AP_{50}, and AP_{75} by 7.4%, 21.0%, and 3.0%, respectively. For the very tiny and tiny-scale objects, the performance is increased by 7.4% in AP_{vt} (0.1% versus 7.5%) and 12.1% in AP_t (7.8% versus 19.9%),

[2]https://github.com/Chasel-Tsui/mmdet-aitod

[3]https://github.com/shaunyuan22/SODA-mmrotate

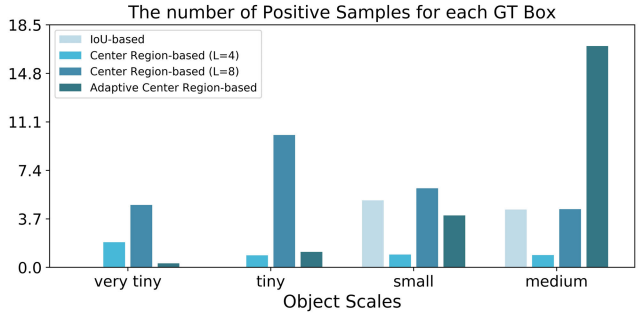


Fig. 4. Analysis of the number of positive samples assigned to each ground-truth box in different scale ranges.

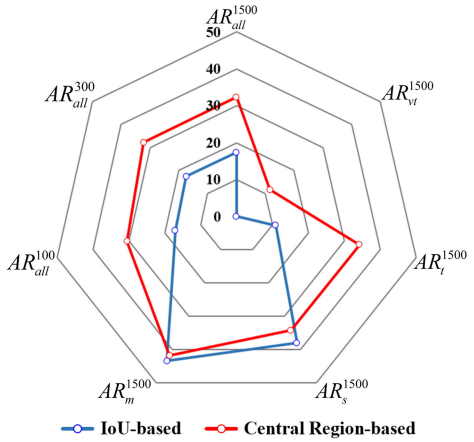


Fig. 5. Visualization of average recall (AR) improvements on the AI-TOD test set by applying the proposed CR label assignment to RPN.

which confirms the effectiveness of our proposed method for tiny object detection. Besides, we analyze the number of positive samples assigned to each ground-truth box under the IoU-based label assignment and the proposed CR label assignment in Fig. 4. We can see that the IoU-based label assignment tends to assign more positive samples to small- and medium-scale objects while it fails to assign positive examples to very tiny-scale objects. In contrast, our proposed CR label assignment increases the number of positive samples for very tiny and tiny objects. Benefiting from the above advantages, the proposed CR label assignment makes the RPN obtain 14.9% AR^{1500}_{all}, 11.7% AR^{1500}_{vt}, and 23.4% AR^{1500}_{t} improvements over the IoU-based label assignment, as shown in Fig. 5. However, the increased number of positive samples for very tiny and tiny-scale objects in CR label assignment harms the number of positive samples for small- and medium-scale objects to a certain extent, which results in the performance degradation in AP_s and AP_m.

a) Analysis of the CR areas: Table II reports the impact of different CR areas in the CR label assignment. When we set the CR with a large area, the CR label assignment tends to assign more positive samples for small- and medium-scale objects, which overwhelms the positive samples for very tiny and tiny-scale objects and degrades their detection performance. Conversely, a small CR area limits the number of positive samples for objects and hinders the detection performance in each scale range. Based on the above analysis, we set the L to 8 in the CR label assignment for the later

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TABLE II

ABLATION STUDIES OF THE PROPOSED CR LABEL ASSIGNMENT WITH DIFFERENT CR AREAS ON THE AI-TOD TEST SET. $R = 0.2$ INDICATES THE RATIO OF THE CR AREA TO THE BOX

L	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
4	18.1	46.0	10.5	6.5	17.6	23.0	30.0
8	**19.0**	**47.9**	**11.3**	**7.5**	**19.9**	22.8	29.0
16	18.6	46.3	11.3	7.3	18.9	24.3	30.1
32	18.1	45.6	11.0	5.2	17.4	24.2	32.2
R = 0.2	17.9	45.0	10.9	3.0	16.9	**25.2**	**33.3**

TABLE III

COMPARISON RESULTS OF PROPOSED CR LABEL ASSIGNMENT WITH SIMILAR METHODS ON THE AI-TOD TEST SET. † MEANS WITH CENTER SAMPLING

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
RPN [22]	11.6	26.9	8.3	0.1	7.8	**24.4**	**34.1**
GA-RPN [61]	15.4	39.5	9.1	1.4	15.1	22.0	29.4
RPN + FCOS [63]	17.2	43.1	10.3	3.8	15.8	23.8	32.2
RPN + FCOS† [63]	18.0	45.1	10.6	4.4	17.4	23.4	30.4
Cascade-RPN [62]	18.3	44.3	**11.9**	5.2	17.9	23.6	32.0
ATSS + RPN [50]	18.4	45.7	11.0	5.2	18.2	23.8	31.3
CR+RPN (Ours)	**19.0**	**47.9**	11.3	**7.5**	**19.9**	22.8	29.0

experiments. Further, we also report the result with an adaptive CR approach in the last row of Table II. The adaptive CR approach defines the CR based on the scale of the ground-truth box, which leads to more positive samples for larger scale objects while limiting the number of positive samples for tiny-scale objects, as shown in Fig. 4. This property makes it still struggles with very tiny-scale object detection, where AP_{vt} is only 3.0%.

b) Comparisons with similar methods: Table III reports the comparison results of our proposed CR label assignment with similar methods, and our method achieves better performance. GA-RPN [61] utilizes semantic features to guide the anchor generation and achieves certain performance improvements. While the IoU-based label assignment in GA-RPN still hinders the performance on very tiny and tiny-scale objects (1.4% in AP_{vt} and 15.1% in AP_{vt}). As shown in the third row of Table III, Cascade-RPN [62] demonstrates its effectiveness in tiny object detection. This is mainly because the cascade-RPN employs a region-based label assignment in the first phase, increasing the number of positive samples for tiny objects. However, the CR ratio (0.2) in cascade-RPN is too small to obtain sufficient positive samples for objects in very tiny and tiny scales. In contrast, our proposed CR label assignment set all anchors falling inside objects with a long edge less than 8 as positive samples, providing more positive samples for very tiny and tiny-scale objects. Thus, compared to Cascade-RPN, our method has a 2.3% and 2.0% improvement in AP_{vt} and AP_t. Besides, our approach yields a 0.6% AP improvement over the ATSS [50].

2) Effectiveness of Gated Context Aggregation: To enhance the feature representation of tiny objects, we extend the FPN with the proposed GCA module to aggregate the valuable context information for tiny objects. As depicted in the third row of Table I, the GCA module leads to a 0.4% improvement

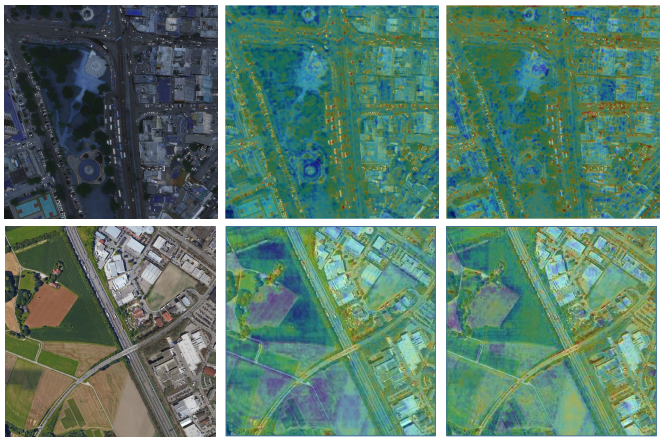


Fig. 6. Visualization of the feature map at the P_2 level in FPN without and with the GCA module.

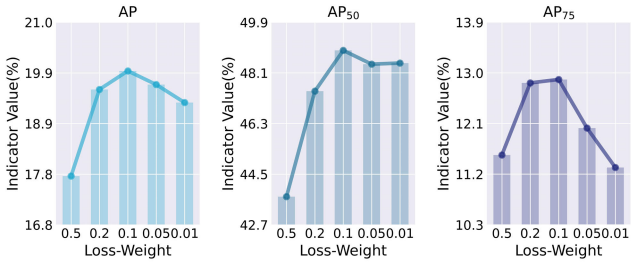


Fig. 7. Ablation study of the loss weight of the generated RoI features, where the loss weight is set to [0.5, 0.2, 0.1, 0.05, 0.01], and the three plots correspond to the values of AP, AP_{50}, and AP_{75} on the AI-TOD test set.

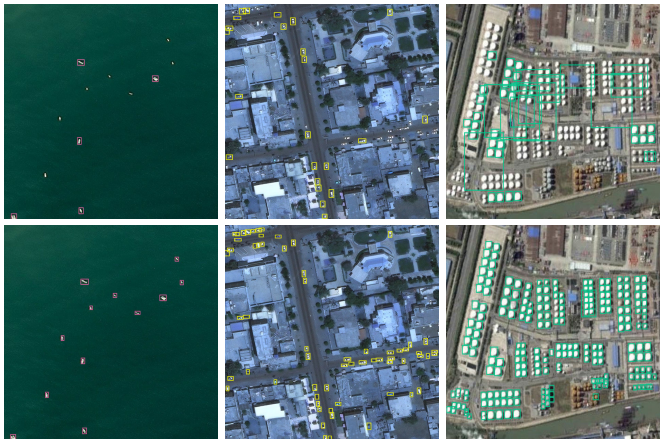


Fig. 8. (Top) Visualization of the detection results using the baseline and (bottom) our proposed MENet on the AI-TOD test set.

in AP. Additionally, Fig. 6 provides the visualization of the feature map with and without the GCA module. We can see that the GCA module enhances the feature representation of tiny objects in complex scenes.

3) Effectiveness of pRoI Features Generator: In the two-stage anchor-based detectors, the IoU-sensitive problem for tiny objects also harms the label assignment in the R-CNN. Thus, we design a pRoI generator to generate a rich diversity of pRoI features for tiny objects in the R-CNN. As shown in Table I, when employing the proposed pRoI generator, the detection performance improves further in different scale

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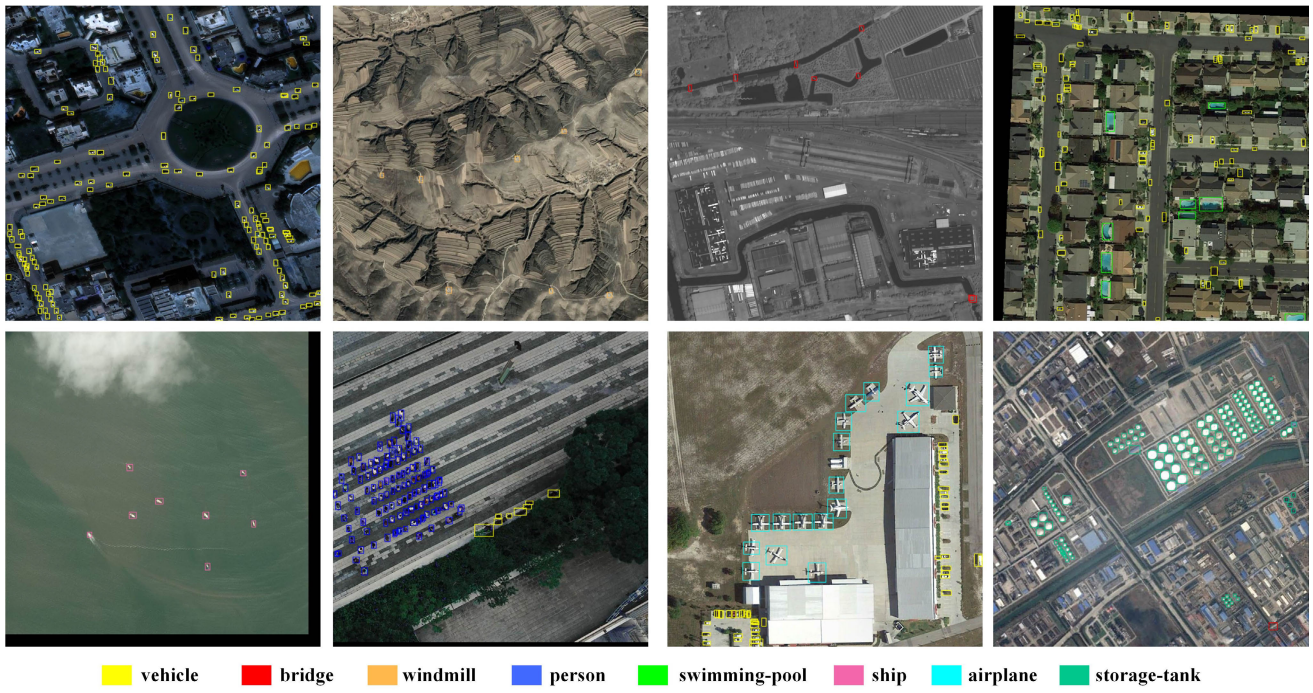


Fig. 9. Visualization of detection results with our proposed MENet on the AI-TOD test set, where one color represents one class. The proposed MENet precisely detects tiny objects in both dense and sparse scenes.

TABLE IV
MODEL COMPLEXITY COMPARISON BETWEEN OUR PROPOSED MENET AND BASELINE METHOD

Method	AP	AP_{50}	AP_{75}	FPS	# Parmas	FLOPs
Faster R-CNN	11.6	26.9	8.3	10.7	41.16M	134.42G
Ours	20.4	50.0	12.9	9.6	41.95M	148.25G

TABLE V
GENERALIZATION ANALYSIS OF OUR PROPOSED METHOD APPLIED TO DIFFERENT DETECTORS ON THE AI-TOD TEST SET. RESNET-50 WITH FPN IS CHOSEN AS THE BACKBONE

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Faster R-CNN [22]	11.6	26.9	8.3	0.1	7.8	24.4	34.1
+ Ours	20.4	50.0	12.9	8.9	21.4	23.2	31.0
Cascade R-CNN [55]	13.7	30.5	10.4	0.0	9.9	26.1	36.4
+ Ours	22.0	50.9	15.7	9.6	22.9	25.3	33.8
DetectoRS [65]	14.7	32.1	11.7	0.1	10.3	28.5	37.9
+ Ours	23.5	53.6	17.0	11.5	24.5	27.0	35.5

objects, where the improvement is more obvious in AP_{vt} (7.5% versus 8.4%).

a) Analysis of the loss weight: Fig. 7 shows the ablation study of the loss weight of the generated RoI features. It can be observed that a larger loss weight deteriorates the detection performance. This performance degradation is mainly attributed to the over-focus on the generated pRoI features, making the R-CNN biased toward the positive classes and prone to detecting false positive results. On the other hand, a smaller loss weight limits the benefits of the proposed pRoI generator. Thus, we set the loss weight to 0.1 as the default setting.

4) Effectiveness of MENet: With the joint extension of the proposed CR label assignment, GCA module, and pRoI generator to the baseline, we get our proposed MENet and obtain a further promotion in detection performance. As depicted in the last row of Table I, our MENet achieves the best AP of 20.4%, which demonstrates the complementary of our proposed modules. Besides the above quantitative analysis, we also visualize the comparative detection results in Fig. 8. Obviously, our proposed method effectively alleviates the miss-detection problem for tiny objects and achieves better detection performance than the baseline.

a) Model complexity: Table IV reports the model complexity of our proposed MENet. Since the proposed CR label assignment and pRoI generator are only introduced in the training phase and can be discarded in the inference phase, these modules do not increase the model complexity in the inference phase. As for the GCA module, it only introduces a small number of convolutional layers and leads to a negligible increase in the number of model parameters (0.79M). Besides, compared with the secondary computational complexity of self-attention [51], the proposed GCA module has the complexity of O(N), which brings a slight burden in model complexity (134.42 GFLOPs versus 148.25 GFLOPs).

b) Generalization analysis of different detectors: As reported in Table V, our proposed method endows these detectors with the ability to detect tiny objects and improve the AP of faster R-CNN [22], Cascade R-CNN [55], and DetectoRS [65] by 8.8%, 8.3%, and 8.8%, respectively. Besides, we find the performance improvement is more prominent for objects in the very tiny and tiny scales, confirming the effectiveness of our proposed method in tiny object detection once more.

c) Generalization analysis of different backbones: As shown in Table VI, when we employ the HRNet-W32 [64]

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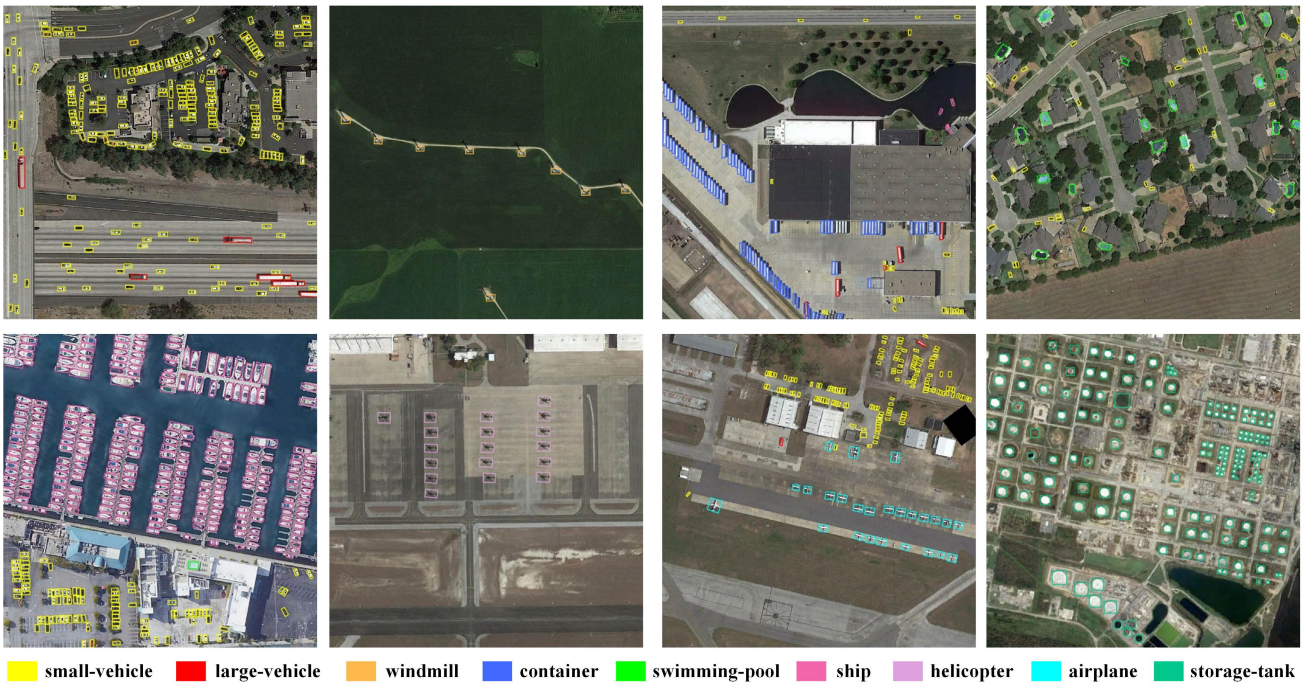


Fig. 10. Visualization of detection results with our proposed MENet on the SODA-A test set, where one color represents one class. The proposed MENet also works well in the scene with arbitrarily rotated objects.

TABLE VI
COMPARISONS PERFORMANCE ON THE AI-TOD TEST SET. WE REPORT THE CLASS-WISE DETECTION RESULTS WITH AP

Model	Backbone	AI	BR	ST	SH	SP	VE	PE	WM	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
RetinaNet [66]	ResNet-50	0.01	6.62	1.84	20.87	0.06	5.67	1.75	0.53	4.7	13.6	2.1	2.0	5.4	6.3	7.6
SSD-512 [24]	VGG-16	14.52	3.13	10.89	13.05	1.92	7.84	3.12	1.48	7.0	21.7	2.8	1.0	5.4	11.5	13.5
Faster R-CNN [22]	ResNet-50	22.51	5.07	19.48	19.19	8.99	13.10	4.58	0.04	11.6	26.9	8.3	0.0	7.8	24.4	34.1
Cascade R-CNN [55]	ResNet-50	25.17	7.09	22.68	24.23	9.72	15.18	5.35	0.04	13.7	30.5	10.4	0.0	9.9	26.1	36.4
FCOS [63]	ResNet-50	19.28	7.37	26.05	27.40	5.77	17.37	5.63	2.56	13.9	35.5	8.6	2.7	12.0	20.2	32.2
ATSS [50]	ResNet-50	23.32	5.37	26.99	26.56	4.44	19.12	5.75	0.57	14.0	33.8	9.2	2.2	12.2	21.5	31.9
M-CenterNet [57]	DLA-34	18.59	10.58	27.55	22.27	7.53	18.60	9.17	2.03	14.5	40.7	6.4	6.1	15.0	19.4	20.4
DetectoRS [65]	ResNet-50	26.39	9.37	23.13	25.96	11.03	14.92	5.93	0.80	14.7	32.1	11.7	0.1	10.3	28.5	37.9
FSANet [11]	ResNet-50	22.21	10.81	28.24	29.84	9.40	19.57	5.89	4.23	16.3	41.4	9.8	4.4	14.6	23.4	33.3
FSANet [11]	Swin-T	27.67	8.79	33.41	31.64	15.34	23.80	8.02	3.28	19.0	47.2	11.6	4.7	17.5	26.2	36.9
MAV23 [67]	Swin-T	24.62	13.47	28.55	26.35	13.70	18.35	8.13	4.39	17.2	47.7	8.0	8.9	18.1	21.2	28.4
ADAS-GPM [12]	ResNet-50	-	-	-	-	-	-	-	-	20.1	49.7	12.4	7.4	19.8	24.9	32.1
SAFF-SSD [68]	2L-Transformer	-	-	-	-	-	-	-	-	21.1	49.9	14.9	7.0	20.8	**30.1**	**38.8**
NWD [10]	ResNet-50	24.35	12.24	35.15	34.43	10.10	24.18	9.50	3.52	19.2	48.5	11.1	7.6	19.0	23.9	31.6
NWD [10]	Swin-T	26.96	15.49	**38.20**	36.04	17.64	26.05	10.80	6.10	22.2	55.2	13.4	8.6	22.1	27.2	35.3
MENet	ResNet-50	22.61	15.98	35.15	39.81	10.80	22.71	9.26	6.58	20.4	50.0	12.9	8.9	21.4	23.2	31.0
MENet	HRNet-w32	22.48	**19.69**	35.82	**42.26**	14.17	**26.40**	**11.16**	7.07	22.4	53.5	14.6	9.6	23.5	25.8	31.8
MENet	Swin-T	**27.52**	16.44	37.36	42.11	**18.77**	24.93	10.19	**8.16**	**23.2**	**56.2**	**15.0**	**9.7**	**23.9**	25.3	34.4

as the backbone, the detection performance achieves a 2.0% AP further improvement. We attribute this performance gain to the fact that HRNet contains high-resolution features throughout the network that better preserve the feature representation for tiny objects. Furthermore, when integrating the Swin-T [53] into the proposed MENet, the detection performance achieves the best AP of 23.2%, demonstrating the remarkable robustness and effectiveness of our proposed method.

E. Comparison With the State-of-the-Art

1) Results on AI-TOD Dataset: We report the comparison performance on the AI-TOD dataset in Table VI. As shown in Table VI, our proposed MENet achieves a performance of 20.4% AP and outperforms other methods, indicating the effectiveness of our MENet in tiny object detection. In comparison with the anchor-based methods [22], [55], [65], MENet obtains more than 5.0% AP performance gain. The IoU-based label assignment in these methods

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TABLE VII
COMPARISONS PERFORMANCE ON THE SODA-A TEST SET. WE REPORT THE CLASS-WISE DETECTION RESULTS WITH AP_S

Model	Backbone	AI	HC	SV	LV	SH	CO	ST	SP	WM	AP_S	AP_{eS}	AP_{rS}	AP_{gS}
R-RetinaNet [66]	ResNet-50	33.20	18.11	23.21	4.46	23.18	10.57	28.61	27.78	16.81	20.7	6.1	16.6	26.4
CSL [69]	ResNet-50	33.07	19.06	23.04	3.69	23.85	10.02	28.53	27.93	16.79	20.7	6.3	16.6	27.0
KLD [70]	ResNet-50	33.16	16.12	23.47	5.38	25.79	15.77	29.21	28.78	16.90	21.6	6.5	17.1	28.5
R^3Det [29]	ResNet-50	40.33	17.67	15.65	9.45	17.24	19.51	33.08	30.78	19.27	22.6	4.4	16.9	32.2
S^2ANet [30]	ResNet-50	36.23	20.01	28.32	8.31	30.17	19.71	28.47	30.44	18.18	24.4	8.2	19.7	30.6
R-Faster R-CNN [22]	ResNet-50	44.04	22.12	29.53	12.72	35.98	23.38	39.91	31.71	19.47	28.8	9.2	23.9	36.8
Oriented R-CNN [71]	ResNet-50	45.36	22.29	29.88	17.13	37.35	28.33	39.86	34.17	19.95	30.5	9.4	25.1	39.4
RoI Transformer [28]	ResNet-50	48.25	22.69	32.34	16.40	40.17	31.21	41.31	34.73	20.33	31.9	10.5	26.7	41.4
MENet	ResNet-50	47.68	23.80	31.47	18.07	39.52	32.68	41.55	33.17	20.70	32.1	10.8	26.3	41.4
RoI Transformer + Ours	ResNet-50	**50.42**	**26.36**	**32.61**	**21.48**	**41.22**	**36.58**	**41.98**	**34.82**	**20.86**	**34.0**	**11.3**	**28.1**	**43.0**

severely limits their performance in tiny object detection. Likewise, our proposed MENet exceeds the anchor-free methods ATSS [50] and FSANet [11] by a considerable margin in detection performance. Compared with the recent tiny object detection methods, our proposed method outperforms NWD [10] by 1.2% AP and ADAS-GPM [12] by 0.3% AP. We attribute this performance improvement to the fact that our MENet enhances the tiny objects from different stages (i.e., RPN, FPN, and R-CNN) of the detector, which provides a more comprehensive enhancement for tiny objects than previous work. Our proposed MENet also shows a more significant performance improvement in AP_{75}. This is mainly because the proposed pRoI generator can generate rich high-quality RoI features, thus improving the capability in high-quality localization of our MENet. Compared with the transformer-based methods [67], [68], the MENet with a Swin-T [53] backbone achieves better performance. Besides, our proposed MENet achieves remarkable performance in most categories, as shown in Table VI. Apart from the above qualitative analysis, we also visualize the detection results in Fig. 9. The proposed MENet can detect tiny objects with dense or sparse distribution.

2) Results on SODA-A Dataset: Table VII reports the comparison performance between our proposed method and recent rotated object detection methods [28], [29], [30], [69], [70], [71] on the SODA-A dataset. Since the SODA-A dataset is annotated with rotated bounding boxes, we employ the rotated faster R-CNN [22] as our baseline. As presented in Table VII, our proposed MENet achieves a 3.3%, 1.6%, 2.4%, and 4.6% performance improvement over the baseline in AP_S, AP_{eS}, AP_{rS}, and AP_{gS}, respectively. Besides, our proposed method also performs better than representative rotated detectors RoI-transformer [28] and Oriented R-CNN [71]. Furthermore, when combining our proposed method with a more powerful detector (e.g., RoI-Transformer), the detection performance can be further enhanced and achieves the best AP of 34.0%, as shown in the last row of Table VII. The effectiveness of our proposed method on the SODA-A dataset demonstrates its good generalization ability once again. The detection results are visualized in Fig. 10.

V. CONCLUSION

In this article, we focus on tiny object detection in RSIs and propose a MENet to enhance the tiny objects from different stages of the detector. In particular, we present a CR label assignment to replace the IoU-based label assignment in RPN that can provide sufficient positive samples for tiny objects. Considering the poor-quality feature representations of tiny objects, we design a GCA module to enhance their feature representation by selectively aggregating valuable context information. Additionally, we devise a pRoI generator that directly generates high-quality RoI features of tiny objects in the R-CNN. Extensive experiments on two datasets show the effectiveness of our proposed MENet in tiny object detection. In further research, we will concentrate on tiny object detection in large-scale RSIs (e.g., over 20 000 × 20 000 pixels), which is significant for the practical application of RSOD.

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